ELEC 7410 Solution: Homework Assignment #5 Oct. 21, 2025

1. (**Problem 21, Chapter 8, Gubner**.) Given $U = X + Y$, $V = X - Y$. Fix u and v:

$$x + y = u \text{ and } x - y = v \;\Rightarrow\; \text{ one solution } x_1 = \frac{u+v}{2},\; y_1 = \frac{u-v}{2}$$

$$\text{Jacobian } J(x,y) = \begin{bmatrix} \frac{\partial(x+y)}{\partial x} & \frac{\partial(x+y)}{\partial y} \\ \frac{\partial(x-y)}{\partial x} & \frac{\partial(x-y)}{\partial y} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$\Rightarrow\; \det(J(x_1,y_1)) = -2 \text{ and } |\det(J(x_1,y_1))| = 2 \neq 0$$

$$\text{Hence } f_{UV}(u,v) = \frac{f_{XY}(x_1,y_1)}{|\det(J(x_1,y_1))|} = \frac{1}{2} f_{XY}\big(\frac{u+v}{2}, \frac{u-v}{2}\big)$$

2. (**Problem 23, Chapter 8, Gubner**.) Given

$$f_{XY}(x,y) = f_X(x) f_Y(y) = \Big(\frac{\lambda}{2} e^{-\lambda|x|}\Big)\Big(\frac{\lambda}{2} e^{-\lambda|y|}\Big), \quad U = X,\;\; V = \frac{Y}{X}.$$

Fix u and v:

$$x = u \text{ and } \frac{y}{x} = v \;\Rightarrow\; \text{ one solution } x_1 = u,\; y_1 = v x_1 = v u$$

$$\text{Jacobian } J(x,y) = \begin{bmatrix} \frac{\partial x}{\partial x} & \frac{\partial x}{\partial y} \\ \frac{\partial (y/x)}{\partial x} & \frac{\partial (y/x)}{\partial y} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -\frac{y}{x^2} & \frac{1}{x} \end{bmatrix}$$

$$\Rightarrow\; \det(J(x_1,y_1)) = \frac{1}{x_1} \text{ and } |\det(J(x_1,y_1))| = \frac{1}{|x_1|} = \frac{1}{|u|} \neq 0$$

$$\text{Hence } f_{UV}(u,v) = \frac{f_{XY}(x_1,y_1)}{|\det(J(x_1,y_1))|} = |u| \frac{\lambda^2}{4} e^{-\lambda|u|} e^{-\lambda|uv|}\,, -\infty < u, v < \infty$$

$$\begin{aligned} f_V(v) &= \int_{-\infty}^{\infty} f_{UV}(u,v)\,du \overset{\text{symmetry in u}}{=} 2\int_0^{\infty} \frac{\lambda^2}{4} u e^{-\lambda(u+u|v|)}\,du \\ &= \frac{\lambda^2}{2} \times \frac{1}{\lambda(1+|v|)} E\{U'\} \text{ where } U' \sim \exp(\lambda(1+|v|)) \\ &= \frac{\lambda^2}{2} \times \frac{1}{\lambda^2(1+|v|)^2} = \frac{1}{2(1+|v|)^2}\,, -\infty < v < \infty \end{aligned}$$

3. (**Problem 28, Chapter 8, Gubner**.) Given $X, W \sim \mathcal{N}(0,1)$ and $Y = X^3 + W$. We know that $\hat{A} = \Sigma_{XY}\Sigma_{YY}^{-1}$ where $\Sigma_{XY} = \text{cov}(X,Y) = \text{cov}(\tilde{X},\tilde{Y})$ and $\Sigma_{YY} = \text{cov}(Y,Y) = \text{cov}(\tilde{Y},\tilde{Y})$. Also $\hat{b} = \hat{A} m_Y - m_X$. We have

$$m_Y = E\{X^3\} + E\{W\} = 0 + 0 = 0,\; m_X = E\{X\} = 0 \;\Rightarrow\; \hat{b} = 0$$

For $X \sim \mathcal{N}(m, \sigma^2)$, $E\{(X-m)^{2n}\} = 1 \times 3 \times \cdots \times (2n-3) \times (2n-1)\sigma^{2n}$. Therefore,

$$\begin{aligned}
E\{Y^2\} &= \underbrace{E\{X^6\}}_{=15} + \underbrace{E\{W^2\}}_{=1} + 2E\{X^3\}\underbrace{E\{W\}}_{=0} = 16 \\
\Rightarrow\ \Sigma_{YY} &= E\{Y^2\} = 16 \\
\Sigma_{XY} &= E\{XY\} - m_X m_Y = E\{X^4\} + E\{X\}E\{W\} - 0 = 3 + 0 = 3 \\
\Rightarrow\ \hat{A} &= \frac{3}{16} \text{ and } \hat{X} = \frac{3}{16}Y
\end{aligned}$$

4. (**Problem 29, Chapter 8, Gubner**.) $X \sim \mathcal{N}(0,1), W \sim$ Laplace(λ), i.e., $f_W(w) = (\lambda/2)e^{-\lambda|w|}$. Therefore, $E\{W\} = 0$ and $E\{W^2\} = 2/\lambda^2$. Given $Y = X + W$. We have

$$\begin{aligned}
m_X &= m_W = 0 \ \Rightarrow \hat{b} = 0 \\
E\{Y^2\} &= \underbrace{E\{X^2\}}_{=1} + \underbrace{E\{W^2\}}_{=2/\lambda^2} + 2E\{X\}\underbrace{E\{W\}}_{=0} = 1 + \frac{2}{\lambda^2} \\
\Rightarrow\ \Sigma_{YY} &= E\{Y^2\} = 1 + \frac{2}{\lambda^2}, \\
\Sigma_{XY} &= E\{XY\} - m_X m_Y = E\{X^2\} + E\{X\}E\{W\} - 0 = 1 + 0 = 1 \\
\Rightarrow\ \hat{A} &= \Sigma_{XY}\Sigma_{YY}^{-1} = \frac{\lambda^2}{2+\lambda^2} \text{ and } \hat{X} = \frac{\lambda^2}{2+\lambda^2}Y
\end{aligned}$$

5. (**Problem 3, Chapter 9, Gubner**.) $X \sim \mathcal{N}(0,1)$ and $Y = 3X$.

   (a) By Def. (9.4) of Gubner, X and Y are jointly Gaussian iff $c_1X + c_2Y \sim \mathcal{N}(m, \sigma^2)$ for any c_1 and c_2, not both zero. We have $Z = c_1X + c_2Y = \underbrace{(c_1 + 3c_2)}_{a} X \sim \mathcal{N}(0, a^2)$. Therefore, X and Y are jointly Gaussian.

   (b)
$$\text{cov}\left(\begin{bmatrix} X \\ Y \end{bmatrix} \begin{bmatrix} X & Y \end{bmatrix}\right) = \begin{bmatrix} \Sigma_{XX} & \Sigma_{XY} \\ \Sigma_{YX} & \Sigma_{YY} \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 3 & 9 \end{bmatrix}$$

   (c)
$$f_{XY}(x,y) = f_X(x)\underbrace{f_{Y|X}(y|x)}_{\delta(y-3x)} = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}\delta(y-3x) \ : \quad \text{NOT continuous}$$

6. (**Problem 11, Chapter 9, Gubner**.) We first derive the characteristic function (CF) $\phi_Z(\eta)$ for scalar $Z \sim \mathcal{N}(\mu, \sigma^2)$. We have

$$\phi_Z(\eta) = E\{e^{j\eta Z}\} = \frac{1}{\sqrt{2\pi}\sigma}\int_{-\infty}^{\infty} e^{j\eta z}e^{-(z-\mu)^2/(2\sigma^2)}\,dz = \frac{1}{\sqrt{2\pi}\sigma}\int_{-\infty}^{\infty} e^{-a/2}\,dz$$

where with $b = \mu + j\eta\sigma^2$,

$$a = (z-\mu)^2/\sigma^2 - 2j\eta z = \frac{z^2 - 2bz + b^2 - b^2 + \mu^2}{\sigma^2} = \frac{(z-b)^2}{\sigma^2} + \frac{\mu^2 - b^2}{\sigma^2}\,.$$

Therefore,

$$\phi_Z(\eta) = e^{-(\mu^2-b^2)/(2\sigma^2)} \underbrace{\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-(z-b)^2/(2\sigma^2)}\, dz}_{=1} = e^{j\eta\mu - \eta^2\sigma^2/2}\,. \tag{1}$$

Back to the original problem where we are given $n-$dimensional $\boldsymbol{X}$,

$$\boldsymbol{X} \sim \mathcal{N}(\boldsymbol{m}, \boldsymbol{C}) \;\; \text{with} \;\; \phi_X(\boldsymbol{\nu}) = E\{e^{j\boldsymbol{\nu}^\top \boldsymbol{X}}\} = e^{j\boldsymbol{\nu}^\top \boldsymbol{m} - \boldsymbol{\nu}^\top \boldsymbol{C}\boldsymbol{\nu}/2}$$

Now set $Y = \sum_{i=1}^n a_i X_i$ for some scalars a_i's with $\boldsymbol{a} = [a_1 \; a_2 \; \cdots \; a_n]^\top$. We can express scalar Y as $Y = \boldsymbol{a}^\top \boldsymbol{X}$. Then with $\boldsymbol{\nu} = \eta \boldsymbol{a}$,

$$\phi_Y(\eta) = E\{e^{j\eta Y}\} = E\{e^{j\boldsymbol{\nu}^\top \boldsymbol{X}}\} = e^{j\boldsymbol{\nu}^\top \boldsymbol{m} - \boldsymbol{\nu}^\top \boldsymbol{C}\boldsymbol{\nu}/2} = e^{j\eta \boldsymbol{a}^\top \boldsymbol{m} - \eta^2 \boldsymbol{\nu}^\top \boldsymbol{C}\boldsymbol{\nu}/2} = e^{j\eta\mu - \eta^2\sigma^2/2} \tag{2}$$

where $\mu = \boldsymbol{a}^\top \boldsymbol{m}$ and $\sigma^2 = \boldsymbol{\nu}^\top \boldsymbol{C}\boldsymbol{\nu}$. The CF $\phi_Y(\eta)$ in (2) has the form of the CF in (1).